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Semiconductor package with multilayer bonding pads

Abstract

A semiconductor device includes a substrate that includes an upper protection layer and a plurality of upper bonding pads, a semiconductor chip on the substrate, and a plurality of bonding wires connected to the semiconductor chip and the upper bonding pads. Each of the upper bonding pads includes a first conductive pattern, a second conductive pattern that covers a top surface and a sidewall of the first conductive pattern and includes a metal element the same as a metal element of the first conductive pattern, and a bonding layer on the second conductive pattern. A width at the top surface of the first conductive pattern is less than a width at a bottom surface of the first conductive pattern. The upper protection layer covers sidewalls of the second conductive pattern.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This U.S. nonprovisional application claims priority under 35 U.S.C § 119 to Korean Patent Application No. 10-2021-0161354 filed on Nov. 22, 2021 in the Korean Intellectual Property Office, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

(2) The present inventive concepts relate to a semiconductor package, and more particularly, to a semiconductor package including bonding pads.

(3) A semiconductor package is provided to implement an integrated circuit chip to qualify for use in electronic products. A semiconductor package may be fabricated by mounting a semiconductor chip on a package substrate, and then using bonding wires or bumps to electrically connect the semiconductor chip to the package substrate. For example, a printed circuit board (PCB) may be used as the package substrate. With the development of the electronics industry, various research has been conducted to improve reliability and durability of semiconductor packages.

SUMMARY

(4) Some embodiments of the present inventive concepts provide a compact-sized semiconductor package.

(5) Some embodiments of the present inventive concepts provide a semiconductor package with improved electrical properties and increased reliability.

(6) According to some embodiments of the present inventive concepts, a semiconductor package may comprise: a substrate that includes an upper protection layer and a plurality of upper bonding pads; a semiconductor chip on the substrate; and a plurality of bonding wires connected to the semiconductor chip and the upper bonding pads. Each of the upper bonding pads may include: a first conductive pattern; a second conductive pattern that covers a top surface and a sidewall of the first conductive pattern and includes a metal element the same as a metal element of the first conductive pattern; and a bonding layer on the second conductive pattern. A width at the top surface of the first conductive pattern may be less than a width at a bottom surface of the first conductive pattern. The upper protection layer may cover sidewalls of the second conductive pattern.

(7) According to some embodiments of the present inventive concepts, a semiconductor package may comprise a substrate that includes an upper protection layer and an upper bonding pad. The upper bonding pad may include: a first conductive pattern; a second conductive pattern on the first conductive pattern; and a bonding layer on the second conductive pattern. A width at a top surface of the first conductive pattern may be less than a width at a bottom surface of the first conductive pattern. The second conductive pattern may cover a sidewall and the top surface of the first conductive pattern. A width at a top surface of the second conductive pattern may be about 90% to about 110% of the width at the bottom surface of the first conductive pattern. The upper protection layer may cover sidewalls of the second conductive pattern.

(8) According to some embodiments of the present inventive concepts, a semiconductor package may comprise: a substrate, wherein the substrate includes, a dielectric structure that includes a plurality of stacked dielectric layers, a wiring line between the dielectric layers, a lower pad disposed on a bottom surface of the dielectric structure and electrically connected to the wiring line, an upper bonding pad disposed on a top surface of the dielectric structure and electrically connected to the wiring line, and an upper protection layer on the top surface of the dielectric structure; a solder ball on a bottom surface of the lower pad; a semiconductor chip on a top surface of the substrate; a bonding wire connected to the semiconductor chip and the upper bonding pad; and a molding layer on the substrate, the molding layer covering the semiconductor chip and the bonding wire. The upper bonding pad may include: a first conductive pattern; a second conductive pattern on a top surface of the first conductive pattern, the second conductive pattern including a first metal element the same as a metal element of the first conductive pattern; an intermediate

layer on a top surface of the second conductive pattern, the intermediate layer including a second metal element different from the first metal element; and a bonding layer on a top surface of the intermediate layer, the bonding layer including a third metal element different from the first metal element and different from the second metal element. A width at the top surface of the first conductive pattern may be less than a width at a bottom surface of the first conductive pattern. The upper protection layer may cover a sidewall of the second conductive pattern.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a plan view showing a semiconductor package according to some embodiments.
- (2) FIG. 2A illustrates a cross-sectional view showing a substrate according to some embodiments.
- (3) FIG. 2B illustrates an enlarged view showing section III of FIG. 2A.
- (4) FIG. 2C illustrates a cross-sectional view showing upper bonding pads according to some embodiments.
- (5) FIG. 2D illustrates a cross-sectional view showing upper bonding pads and an upper protection layer according to some embodiments.
- (6) FIG. 2E illustrates a cross-sectional view showing upper bonding pads and an upper protection layer according to some embodiments.
- (7) FIG. 3A illustrates a cross-sectional view taken along I-I' of FIG. 1, showing a semiconductor package according to some embodiments.
- (8) FIG. 3B illustrates a cross-sectional view taken along line II-II' of FIG. 1.
- (9) FIGS. 4A to 4G illustrate cross-sectional views showing a substrate fabrication method according to some embodiments.

DETAIL DESCRIPTION

- (10) In this description, like reference numerals may indicate like components. The following will describe a substrate, a semiconductor package including the same, and a substrate fabrication method according to embodiments of the present inventive concepts.
- (11) FIG. 1 illustrates a plan view showing a semiconductor package according to some embodiments.
- (12) Referring to FIG. 1, a semiconductor package **10** may include a substrate **100**, a semiconductor chip **200**, and bonding wires **300**. The substrate **100** may be a printed circuit board (PCB), but the present inventive concepts are not limited thereto. When viewed in plan view, the substrate **100** may have a central region and an edge region. The edge region of the substrate **100** may be provided between the central region of the substrate **100** and a lateral surface (e.g., four lateral surfaces) of the substrate **100**. The edge region of the substrate **100** may be between lateral side surfaces of the semiconductor chip **200** and respective lateral side surfaces of the substrate **100**. The substrate **100** may include upper bonding pads **150** and an upper protection layer **111**. When viewed in plan view, the upper bonding pads **150** may overlap the edge region of the substrate **100**. The upper protection layer **111** may expose top surfaces of the upper bonding pads **150**. The upper bonding pads **150** may be aligned along a first direction D1 and spaced apart from each other in the first direction D1. The first direction D1 may be parallel to a top surface of the upper protection layer **111**. A second direction D2 may be parallel to the top surface of the upper protection layer **111** and substantially orthogonal to the first direction D1. A third direction D3 may be substantially perpendicular to the top surface of the upper protection layer **111** and substantially vertical to the first and second directions D1 and D2. Terms such as “parallel,” “perpendicular,” “same,” “equal,” “planar,” and “coplanar,” as used herein encompass identity or near identity including variations that may occur, for example, due to manufacturing processes that

fall within allowable tolerances. The term “substantially” may be used herein to emphasize this meaning, unless the context or other statements indicate otherwise. A top surface of the substrate **100** may include the top surfaces of the upper bonding pads **150** and the top surface of the upper protection layer **111**.

(13) The semiconductor chip **200** may be mounted on the top surface of the substrate **100**. For example, when viewed in plan view, the semiconductor chip **200** may overlap the central region of the substrate **100**. In some embodiments, the semiconductor chip **200** may be a memory chip, such as DRAM, SRAM, or NAND Flash. The semiconductor chip **200** may include chip pads **250**. The chip pads **250** may be provided on a top surface of the semiconductor chip **200**. The chip pads **250** may be electrically connected to integrated circuits of the semiconductor chip **200**. The phrase “a certain component is electrically connected to the semiconductor chip **200**” may mean that “the certain component is electrically connected to the integrated circuits through the chip pads **250** of the semiconductor chip **200**.” The expression “two components are electrically connected to each other” may include the meaning that “two components are connected to pass electrical signals from one component to the other either by being directly physically connected to each other or by being indirectly physically connected to each other through other component(s).” The chip pads **250** may include or be formed of a conductive material, such as metal. Though only one chip **200** is shown, in some embodiments, the chip **200** may be part of a stack of chips electrically connected to the substrate **100**.

(14) The bonding wires **300** may be provided on the semiconductor chip **200** and the substrate **100**. The bonding wires **300** may be electrically connected to the chip pads **250** and the upper bonding pads **150**. The semiconductor chip **200** may be electrically connected through the bonding wires **300** to the substrate **100**. The bonding wires **300** may include metal, such as gold (Au).

(15) The following will describe in detail a substrate.

(16) FIG. 2A illustrates a cross-sectional view taken along line I-I' of FIG. 1, showing a substrate according to some embodiments. FIG. 2B illustrates an enlarged view showing section III of FIG. 2A.

(17) Referring to FIGS. 1, 2A, and 2B, the substrate **100** may include a dielectric structure, wiring lines **131**, first conductive vias **135**, second conductive vias **136**, upper bonding pads **150**, a lower protection layer **112**, and lower pads **160**. The dielectric structure may include a plurality of stacked dielectric layers **110**. The dielectric layers **110** may include or be formed of, for example, prepreg. The wiring lines **131** may be provided between the dielectric layers **110**. The wiring line **131** may include or be formed of metal, such as copper. The first and second conductive vias **135** and **136** may penetrate the dielectric layers **110**. The first conductive vias **135** may be provided on top surfaces of the wiring lines **131**, for example, to contact the wiring lines **131** at a top surface of the wiring lines **131**. The second conductive vias **136** may be provided on bottom surfaces of the wiring lines **131**, for example, to contact the wiring lines **131** at a bottom surface of the wiring lines **131**. The first and second conductive vias **135** and **136** may include or be formed of metal, such as copper. It will be understood that when an element is referred to as being “on” or “connected” or “coupled” to another element, it can be directly on or connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, or as “contacting” or “in contact with” another element, there are no intervening elements present at the point of contact. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.).

(18) The upper protection layer **111** may be provided on a top surface of an uppermost one of the dielectric layers **110**. The top surface of the uppermost dielectric layer **110** may be a top surface of the dielectric structure. The upper protection layer **111** may have first holes **119**, or openings, that penetrate therethrough. The first holes **119** may expose the first conductive vias **135**. The upper protection layer **111** may include or be formed of an organic dielectric material, such as a solder

resist material.

(19) The upper bonding pads **150** may be provided on the top surface of the uppermost dielectric layer **110**. The upper bonding pads **150** may be correspondingly provided in the first holes **119**. Each of the upper bonding pads **150** may include a first portion formed in the hole **119** and including a first conductive pattern **151** and a second conductive pattern **152**, and a second portion formed outside of the hole **119** and including an intermediate layer **155** and a bonding layer **157**. The first conductive pattern **151** may be provided on the uppermost dielectric layer **110** and the first conductive via **135** that corresponds to the first conductive pattern **151** (e.g., formed directly on the structure formed of the uppermost dielectric layer **110** and the first conductive via **135**. As illustrated in FIG. 2B, the first conductive pattern **151** may have a first width **W1**, e.g., in the **D1** direction, at a bottom surface thereof. The first width **W1** may be in range from about 17 μm to about 30 μm . Because the first width **W1** is equal to or less than about 30 μm , an electrical short may be prevented between the upper bonding pads **150**. The first conductive pattern **151** may have a trapezoidal shape. For example, the first conductive pattern **151** may have, at a top surface thereof, a width **W11** less than the first width **W1**. A sidewall of the first conductive pattern **151** may be inclined relative to the bottom surface of the first conductive pattern **151**. Though not shown from the direction depicted in FIGS. 2A and 2B, the sidewall may have a curved shape (e.g., a circular shape), from a plan view, or may have four sides connected at edges (e.g., to have a square shape), from a plan view. The first conductive pattern **151** may include or be formed of a first metal element. The first metal element may include or may be, for example, copper. Terms such as “about” or “approximately” may reflect amounts, sizes, orientations, or layouts that vary only in a small relative manner, and/or in a way that does not significantly alter the operation, functionality, or structure of certain elements. For example, a range having ends of different orders of magnitude (e.g., from “about 0.1 to about 1”) may encompass a range such as a 0%-5% deviation around each end (e.g., a 0% to 5% deviation around 0.1 and a 0% to 5% deviation around 1), while a range having ends of the same order of magnitude (e.g., about 15 to about 30) may encompass a range such as a 0%-5% deviation around the smaller of the two ends, particularly where the above deviations maintain the same effect as the listed range.

(20) The second conductive pattern **152** may be provided on the first conductive pattern **151**. For example, the second conductive pattern **152** may cover the top surface and the sidewall of the first conductive pattern **151**. Though not shown from the direction depicted in FIGS. 2A and 2B, the sidewall may have a curved shape, from a plan view, or may have four sides connected at edges, from a plan view. The second conductive pattern **152**, as measured between opposite outer sidewalls in the **D1** direction, may have a uniform width. The second conductive pattern **152** may have the first width **W1**. The first width **W1** may be a width at a top surface **152u** of the second conductive pattern **152**. The second conductive pattern **152** may have a sidewall that is substantially vertical. For example, a substantially right angle may be given as an angle between the sidewall of the second conductive pattern **152** and a bottom surface of the upper protection layer **111**. In this description, the term “vertical” may indicate the meaning of “parallel to the third direction.” The first and second conductive patterns **151** and **152** may be described as first and second conductive components or first and second conductive portions, respectively.

(21) The width at the top surface **152u** of the second conductive pattern **152** may be the same as or similar to the first width **W1**, for example, about 90% to about 110% of the first width **W1**. The width at the top surface **152u** of the second conductive pattern **152** may be in a range, for example, from about 17 μm to about 30 μm . The top surface **152u** of the second conductive pattern **152** may be located at substantially the same level as that of a top surface **111u** of the upper protection layer **111** to be coplanar with the top surface **111u** of the upper protection layer **111**, but the present inventive concepts are not limited thereto. A level of a certain component may indicate a vertical level.

(22) The sidewall of the second conductive pattern **152** may be covered with the upper protection

layer **111**. For example, the sidewall of the second conductive pattern **152** may contact the upper protection layer **111**.

(23) The second conductive pattern **152** may include a first metal element the same as that of the first conductive pattern **151**. According to some embodiments, the second conductive pattern **152** may have a grain different from that of the first conductive pattern **151**, for example, such that a grain boundary is formed between the first conductive pattern **151** and the second conductive pattern **152**. For example, the grain of the second conductive pattern **152** may have a size different from that of the grain of the first conductive pattern **151**. For another example, the grain of the second conductive pattern **152** may have a shape different from that of the grain of the first conductive pattern **151**. For another example, the grain of the second conductive pattern **152** may have a crystal structure different from that of the grain of the first conductive pattern **151**. According to some embodiments, the second conductive pattern **152** may have the same grain as that of the first conductive pattern **151**.

(24) The intermediate layer **155** may be provided on the top surface **152u** of the second conductive pattern **152**. For example, the intermediate layer **155** may cover the top surface **152u** of the second conductive pattern **152**. The intermediate layer **155** may include or be formed of a second metal element. The second metal element may be different from the first metal element. The second metal element may include or be, for example, nickel. A width of the intermediate layer **155** may be, for example, $W2$, which may be greater than $W1$ by, for example, up to about 20%. The width $W2$ may be a width in the $D1$ direction of both a bottom surface and a top surface of the intermediate layer **155**. The intermediate layer **155** may have sidewall surfaces that extend between top and bottom surfaces and are substantially vertical.

(25) The bonding layer **157** may be provided on the intermediate layer **155**. The bonding layer **157** may cover a top surface and sidewalls of the intermediate layer **155**. The top surface of the bonding layer **157** may be a surface to which the bonding wire (see **300** of FIG. **1** or **2A**) is bonded. The bonding layer **157** may have a third width $W3$. The third width $W3$ may be a width at the top surface of the bonding layer **157**. The third width $W3$ may be in a range from about 20 μm to about 33 μm . When the third width $W3$ is less than about 20 μm , the bonding wire **300** may be difficult to bond to the bonding layer **157**. According to some embodiments, because the third width $W3$ is equal to or greater than about 20 μm , the bonding wire **300** may be adequately bonded to the bonding layer **157** even when process errors occur in a process for forming the bonding wire **300**. When the third width $W3$ is greater than about 33 μm , a pitch $P1$ of the upper bonding pads **150** may need to be increased, for example, to avoid short circuits between adjacent upper bonding pads **150**. According to some embodiments, because the third width $W3$ is equal to or less than about 33 μm , the pitch $P1$ of the upper bonding pads **150** may become reduced. Accordingly, the semiconductor package **10** may become high in integration and small in size.

(26) The third width $W3$ may be the same as or greater than the second width $W2$, and may be greater than the first width $W1$. For example, the third width $W3$ may be about 100% to about 120% of the first width $W1$ while being greater than the second width $W2$. If the second width $W2$ and the third width $W3$ are less than the first width $W1$, the second conductive pattern **152** may be exposed and damaged. According to some embodiments, because the third width $W3$ is equal to or greater than about 100% of the first width $W1$, the bonding layer **157** combined with the intermediate layer **155** may completely cover the top surface **152u** of the second conductive pattern **152**. Therefore, the second conductive pattern **152** may be prevented from being damaged. Because the third width $W3$ is equal to or less than about 120% of the first width $W1$, the upper bonding pads **150** may have a fine pitch $P1$.

(27) The bonding layer **157** may have a thickness (e.g., in a vertical direction on the top surface of the intermediate layer **155** and in a horizontal direction on the sidewall surface of the intermediate layer **155**) of about 2 μm to about 5 μm . Because the thickness T of the bonding layer **157** is equal to or greater than about 2 μm , the bonding wire **300** may be adequately bonded to the bonding layer

157. There may be an improvement in bonding reliability between the bonding layer **157** and the bonding wire **300**. Because the thickness T of the bonding layer **157** is equal to or less than about $5\text{ }\mu\text{m}$, the upper bonding pads **150** may have a reduced interval D therebetween, and the upper bonding pads **150** may have a fine pitch $P1$. The bonding layer **157** may have a substantially uniform thickness T . For example, the thickness T of the bonding layer **157** on the top surface of the intermediate layer **155** may be substantially the same as the thickness T of the bonding layer **157** on the sidewalls of the intermediate layer **155**. The thickness T of the bonding layer **157** may have a deviation equal to or less than about $1.5\text{ }\mu\text{m}$. Accordingly, an electrical short may be prevented between the upper bonding pads **150**.

(28) The bonding layer **157** may include or be formed of a third metal element. The third metal element may be different from the first metal element and the second metal element. For example, the third metal element may include or may be gold (Au). In the formation of the bonding layer **157**, the intermediate layer **155** may assist in plating the bonding layer **157**. The intermediate layer **155** may serve as a metal adhesion layer to allow the bonding layer to adequately bond to the second conductive pattern **152**. The bonding layer **157** may further serve as a protection layer. For example, the bonding layer **157** may prevent damage (e.g., oxidation) of the intermediate layer **155** and second conductive pattern **152**.

(29) The pitch $P1$ in the $D1$ direction of the upper bonding pads **150**, or distance between corresponding points of adjacent upper bonding pads **150**, may be a fine pitch. For example, the pitch $P1$ of the upper bonding pads **150** may be in a range of about $40\text{ }\mu\text{m}$ to about $60\text{ }\mu\text{m}$. Because the pitch $P1$ of the upper bonding pads **150** is equal to or greater than about $40\text{ }\mu\text{m}$, an electrical short may be prevented between the upper bonding pads **150** and/or between the bonding wires **300**. Because the pitch $P1$ of the upper bonding pads **150** is equal to or less than about $60\text{ }\mu\text{m}$, the semiconductor package **10** may become small in size and high in integration.

(30) The upper bonding pads **150** may have an interval D , or closest distance between them, of about $10\text{ }\mu\text{m}$ to about $20\text{ }\mu\text{m}$. The interval D between the upper bonding pads **150** may correspond to an interval between the bonding layers **157** that correspond to the upper bonding pads **150**. Because the interval D between the upper bonding pads **150** is equal to or greater than about $10\text{ }\mu\text{m}$, an electrical short may be prevented between the upper bonding pads **150** and/or between the bonding wires **300**. Because the interval D between the upper bonding pads **150** is equal to or greater than about $20\text{ }\mu\text{m}$, the semiconductor package **10** may become small in size and high in integration.

(31) As the first and second conductive patterns **151** and **152** included in one upper bonding pad **150** are provided in corresponding first holes **119**, a plurality of neighboring second conductive patterns **152** may be disposed laterally spaced apart from each other. Accordingly an electrical short may be prevented between the plurality of neighboring second conductive patterns **152**.

(32) When the upper bonding pads **150** do not include the second conductive patterns **152**, and do not include the intermediate layer **155** or bonding layer **157** as shown in FIG. 2B, the first conductive pattern **151** may be required to have an increased size (and the hole **119** may need to be bigger) to satisfy the condition that the third width $W3$ of a bonding layer of upper bonding pads is in a range of about $20\text{ }\mu\text{m}$ to about $33\text{ }\mu\text{m}$. In this case, the first conductive pattern **151** may have an increased width at the bottom surface thereof, and thus an electrical short may occur between the upper bonding pads **150**. Alternatively, there may need to be an increase in the pitch $P1$ of the upper bonding pads **150**. According to some embodiments, because the upper bonding pads **150** include the second conductive patterns **152**, the first conductive pattern **151** may have a reduced size, and the second width $W2$ may be relatively small. Therefore, an electric short may be prevented between the upper bonding pads **150**, and the interval D may be reduced between the upper bonding pads **150**. Accordingly, the pitch $P1$ may be reduced between the upper bonding pads **150**. For brevity of description, the following will discuss a single second conductive pattern **152**.

(33) Referring back to FIG. 2A, the lower protection layer **112** may be provided on a bottom surface of a lowermost one of the dielectric layers **110**. The bottom surface of the lowermost dielectric layer **110** may be a bottom surface of the dielectric structure. The lower protection layer **112** may have second holes **129** that penetrate therethrough. The second holes **129** may expose second conductive vias **136**. The lower protection layer **112** may include or be formed of a dielectric material, such as a solder resist material.

(34) The lower pads **160** may be provided on the top surface of the lowermost dielectric layer **110**. The lower pads **160** may be provided in the second hole **129**. Each of the lower pads **160** may include a lower conductive pattern **161**, a lower intermediate layer **165**, and a lower bonding layer **167**. The lower conductive pattern **161** may be provided on the bottom surface of the lowermost dielectric layer **110** and a bottom surface of the second conductive via **136** that corresponds to the lower conductive pattern **161**. A shape of the second conductive pattern **152** may be variously changed. The second conductive pattern **152** may include or be formed of the first metal element.

(35) The lower intermediate layer **165** may be provided on a bottom surface of the lower conductive pattern **161**. The lower intermediate layer **165** may have a width the same as or less than that of the lower conductive pattern **161**. The lower intermediate layer **165** may include or be formed of the second metal element.

(36) The lower bonding layer **167** may be provided on the bottom surface of the lower intermediate layer **165**. The lower bonding layer **167** may have a bottom surface to which a solder ball (see **600** of FIG. 3A) is bonded. The lower bonding layer **167** may include or be formed of the third metal element. The lower protection layer **112** may cover a sidewall of the lower conductive pattern **161**, a sidewall of the lower intermediate layer **165**, and a sidewall of the lower bonding layer **167**.

(37) The lower pads **160** may have a pitch **P2** greater than the pitch **P1** of the upper bonding pads **150**.

(38) Alternatively, each of the lower pads **160** may include neither the lower intermediate layer **165** nor the lower bonding layer **167**. In this case, each of solder balls (see **600** of FIG. 3A) may be directly disposed on the bottom surface of a corresponding lower conductive pattern **161**.

(39) The lower pads **160** may be electrically connected to the upper bonding pads **150** through the second conductive vias **136**, the wiring lines **131**, and the first conductive vias **135**.

(40) FIG. 2C illustrates an enlarged view of section III depicted in FIG. 2A, showing upper bonding pads according to some embodiments.

(41) Referring to FIG. 2C, each of the upper bonding pads **150** may include a seed pattern **153** in addition to the first conductive pattern **151**, the second conductive pattern **152**, the intermediate layer **155**, and the bonding layer **157**. The seed pattern **153** may be interposed between the first conductive pattern **151** and the second conductive pattern **152**. The seed pattern **153** may cover the top surface and the sidewalls of the first conductive pattern **151**. The second conductive pattern **152** may be formed by a plating process in which the seed pattern **153** is used as an electrode. The seed pattern **153** may include or be formed of metal the same as the first metal element. Alternatively, the seed pattern **153** may include metal different from the first metal element. For example, the seed pattern **153** may include or may be titanium, copper, or any alloy thereof.

(42) FIGS. 2D and 2E illustrate an enlarged view of section III depicted in FIG. 2A, showing upper bonding pads and an upper protection layer according to some embodiments. A duplicate description will be omitted below.

(43) Referring to FIG. 2D, the top surface **152u** of the second conductive pattern **152** may be located at a different level from that of the top surface **111u** of the upper protection layer **111**. For example, the top surface **152u** of the second conductive pattern **152** may be located at a lower level than that of the top surface **111u** of the upper protection layer **111**. The intermediate layer **155** may be provided in an upper portion of the second hole **129**.

(44) Referring to FIG. 2E, the top surface **152u** of the second conductive pattern **152** may be located at a higher level than that of the top surface **111u** of the upper protection layer **111**.

(45) The following will now describe a semiconductor package according to some embodiments.
(46) FIG. 3A illustrates a cross-sectional view taken along I-I' of FIG. 1, showing a semiconductor package according to some embodiments. FIG. 3B illustrates a cross-sectional view taken along line II-II' of FIG. 1.

(47) Referring to FIGS. 1, 3A, and 3B, a semiconductor package **10** may include a substrate **100**, solder balls **600**, a semiconductor chip **200**, bonding wires **300**, and a molding layer **400**. The substrate **100** may be substantially the same as that discussed in the examples of FIGS. 2A to 2E.

(48) The solder balls **600** may be provided on a bottom surface of the substrate **100**. For example, the solder balls **600** may be correspondingly provided on bottom surfaces of the lower pads **160**. The solder balls **600** may include or be formed of metal, such as a solder material. The solder material may include or be one or more of tin (Sn), silver (Ag), zinc (Zn), and any alloy thereof. The substrate **100** may be electrically connected through the solder balls **600** to an external apparatus. The solder balls **600** therefore may be described as external connection terminals.

(49) As illustrated in FIGS. 1 and 3B, the semiconductor chip **200** may be mounted on a top surface of the substrate **100**. The semiconductor chip **200** may include chip pads **250** on the top surface thereof.

(50) According to some embodiments, the bonding wires **300** may be provided on a top surface of the semiconductor chip **200** to electrically connect to the chip pads **250**. The bonding wires **300** may be bonded to top surfaces of the upper bonding pads **150**. For example, the bonding wire **300** may be correspondingly coupled to the bonding layer **157**. As discussed above, because the third width $W3$ is equal to or greater than about $20\text{ }\mu\text{m}$, the bonding wire **300** may be adequately bonded to the bonding layer **157** even when process errors occur in a process for forming the bonding wire **300**.

(51) It is illustrated that the bonding wires **300** are provided on opposite sidewalls of the semiconductor chip **200**, but alternately the bonding wires **300** may be provided on only one sidewall of the semiconductor chip **200**. In at least this manner, aspects of the invention may apply to chip stacks including at least a first chip and one or more chips stacked thereon, for example in a step-wise or zig-zag format. For another example, the bonding wires **300** may be provided on at least three sidewalls of the semiconductor chip **200**.

(52) As shown in FIG. 3B, and referring back to FIG. 2A, according to some embodiments, upper bonding pads **150** may extend vertically from a lower surface of a single, uppermost insulating layer of a substrate **100** (e.g., a lower surface of upper protection layer **111**) through the uppermost insulating layer (e.g., upper protection layer **111**). The bottom surface of the bonding pads **150** may have a surface coplanar with a bottom surface of the uppermost insulating layer of the substrate **100** (e.g., upper protection layer **111**). The top surface of the bonding pads **150** may have a surface above a top surface of the uppermost insulating layer of the substrate **100**. Each upper bonding pad may have a flat top surface and a flat bottom surface. The upper bonding pads **150** may be upper connection terminals, directly adjacent to and contacting bonding wires **300** that connect the substrate **100** to the semiconductor chip **200**.

(53) An adhesion layer (not shown) may further be interposed between the semiconductor chip **200** and the substrate **100**. The adhesion layer may include or may be a die attach film (DAF).

(54) The molding layer **400** may be provided on the top surface of the substrate **100** to cover the semiconductor chip **200** and the bonding wires **300**. The molding layer **400** may include a dielectric polymer, such as an epoxy-based polymer.

(55) The following will describe a substrate fabrication method according to some embodiments.

(56) FIGS. 4A to 4G illustrate cross-sectional views taken along line I-I' of FIG. 1, showing a substrate fabrication method according to some embodiments.

(57) Referring to FIG. 4A, a temporary substrate **900** may be provided. The temporary substrate **900** may include a detachable core. The temporary substrate **900** may have a top surface and a bottom surface that are opposite to each other.

(58) Second conductive layers **161Z** may be formed on the top and bottom surfaces of the temporary substrate **900**. Dielectric layers **110** may be correspondingly formed on and cover the second conductive layers **161Z**. The dielectric layers **110** may be first dielectric layers **110A**. Second conductive vias **136** may be formed in the first dielectric layers **110A** to electrically connect to the second conductive layers **161Z**. Wiring lines **131** may be correspondingly formed on the second conductive vias **136** and the first dielectric layers **110A**. The wiring lines **131** may be spaced apart from each other. The wiring lines **131** may be correspondingly coupled to the second conductive vias **136**.

(59) Referring to FIG. 4B, the formation of the dielectric layers **110** may be repeatedly performed to form second dielectric layers **110B**. The second dielectric layers **110B** may be formed on and cover the first dielectric layers **110A** and the wiring lines **131**. First conductive vias **135** may be formed in the second dielectric layers **110B** and on the wiring lines **131**. The first conductive vias **135** may be electrically connected to the wiring lines **131**.

(60) First conductive layers **151Z** may be formed on the second dielectric layers **110B** and the first conductive vias **135**. The formation of the first conductive layers **151Z** may include performing a plating process. The plating process may include an electroplating process or a chemical plating process. The chemical plating process may include an electroless plating process. The first conductive layers **151Z** may be coupled to the first conductive vias **135**. Therefore, preliminary substrates **20** may be fabricated. The preliminary substrates **20** may be correspondingly formed on the top and bottom surfaces of the temporary substrate **900**. Each of the preliminary substrates **20** may include a corresponding first dielectric layer **110A**, a corresponding second dielectric layer **110B**, a corresponding second conductive layer **161Z**, first conductive vias **135**, a corresponding first conductive layer **151Z**, and second conductive vias **136**.

(61) Differently from the discussion of FIGS. 4A and 4B, the first conductive layers **151Z** may be correspondingly formed on top and bottom surfaces of the temporary substrate **900**. Afterwards, a plurality of second dielectric layers **110B**, a plurality of first conductive vias **135**, a plurality of first dielectric layers **110A**, a plurality of second conductive vias **136**, a plurality of first dielectric layers **110A**, and a plurality of second conductive layers **161Z** may be sequentially formed to form the preliminary substrates **20**.

(62) Referring to FIG. 4C, the preliminary substrates **20** may be separated from the temporary substrate **900**.

(63) Referring to FIG. 4D, one of the separated preliminary substrates **20** is shown.

(64) Referring to FIG. 4E, using one of the separated preliminary substrates **20** as an example, the first conductive layer **151Z** may undergo a patterning process, and the second conductive layer **161Z** may undergo a patterning process. The patterning process of the first conductive layer **151Z** and the patterning process of the second conductive layer **161Z** may be performed in a single or multiple steps.

(65) The first conductive layer **151Z** may be patterned to form a plurality of first conductive patterns **151**. The patterning process of the first conductive layer **151Z** may include an etching process. The patterning process of the first conductive layer **151Z** may further include exposure and development processes. The etching process may include a wet etching process. An upper portion of each of the first conductive patterns **151** may be exposed to the etching process earlier than a lower portion of each of the first conductive patterns **151**, and thus each of the first conductive patterns **151** may have a trapezoidal shape. Therefore, each of the first conductive patterns **151** may have a width **W11** at its top surface less than a first width **W1** at its bottom surface. The first conductive patterns **151** may be correspondingly coupled to the first conductive vias **135**. The first conductive patterns **151** may be laterally spaced apart and electrically separated from each other.

(66) The second conductive layer **161Z** may be patterned to form a plurality of lower conductive patterns **161**. The patterning process of the second conductive layer **161Z** may include an etching

process. The patterning process of the second conductive layer **161Z** may further include exposure and development processes. The etching process may include a wet etching process. Differently from that shown, each of the lower conductive patterns **161** may have a width at its top surface greater than a width at its bottom surface. The lower conductive patterns **161** may be correspondingly coupled to the second conductive vias **136**. The lower conductive patterns **161** may be laterally spaced apart and electrically separated from each other.

(67) Referring to FIG. **4F**, an upper protection layer **111** may be formed on a top surface of an uppermost dielectric layer **110**. The formation of the upper protection layer **111** may include coating a solder resist material. First holes **119** may be formed in the upper protection layer **111** to expose the first conductive patterns **151**.

(68) A lower protection layer **112** may be formed on a bottom surface of a lowermost dielectric layer **110**. The formation of the lower protection layer **112** may include coating a solder resist material. Second holes **129** may be formed on the lower protection layer **112** to expose the lower conductive patterns **161**.

(69) Referring to FIG. **4G**, the first conductive patterns **151** may undergo a plating process to form a plurality of second conductive patterns **152**. The second conductive patterns **152** may cover top and lateral surfaces of the first conductive patterns **151**. The plating process may include an electroplating process or a chemical plating process.

(70) The lower conductive patterns **161** may not be exposed to the plating process. For another example, a plating process may further be processed on the lower conductive patterns **161**. In this case, additional lower conductive patterns (not shown) may further be formed on bottom surfaces of the lower conductive patterns **161**.

(71) Referring back to FIG. **2A**, a plurality of intermediate layers **155** may be correspondingly formed on the second conductive patterns **152**. The intermediate layers **155** may be laterally spaced apart from each other. A plurality of bonding layers **157** may be formed on the intermediate layers **155** to cover top surfaces and sidewalls of the intermediate layers **155**. A plating process may be performed to form the bonding layers **157**. The bonding layers **157** may be disposed laterally spaced apart from each other. Therefore, upper bonding pads **150** may be fabricated. The upper bonding pads **150** may include the first conductive patterns **151**, the second conductive patterns **152**, the intermediate layers **155**, and the bonding layers **157**.

(72) A plurality of lower intermediate layers **165** may be formed on the lower conductive patterns **161** and in the second holes **129**. The lower intermediate layers **165** and the intermediate layers **155** may be formed in a single process or in individual processes. A plurality of lower bonding layers **167** may be formed on the lower intermediate layers **165** and in the second holes **129**. The lower bonding layers **167** and the bonding layers **157** may be formed in a single process or in individual processes. Therefore, lower pads **160** may be fabricated. The lower pads **160** may include the lower conductive patterns **161**, the lower intermediate layers **165**, and the lower bonding layers **167**.

Through the aforementioned exemplary processes, a substrate **100** may be eventually fabricated.

(73) Subsequently, as shown in FIG. **3B**, a semiconductor chip **200** may be disposed on the substrate **100**, and bonding wires may be connected to the semiconductor chip **200** via chip pads **250** and may be connected to the substrate **100** via the upper bonding pads **150**. The bonding of wires to the pads may be performed using known bonding techniques, for example, using pressure, heat, ultrasonic energy, etc. Additional steps may be performed, as described previously, to result in the semiconductor package **10** shown in FIG. **3B**.

(74) According to the present inventive concepts, upper bonding pads may each include a first conductive pattern, a second conductive pattern, an intermediate layer, and a bonding layer. The presence of the second conductive pattern may cause the upper bonding pads to have a fine pitch. An upper protection layer may cover a sidewall of the second conductive pattern to prevent an electrical short between the upper bonding pads. Bonding wires may be prevented from an electrical short, and may each be adequately bonded to a corresponding bonding layer.

(75) This detailed description of the present inventive concepts should not be construed as limited to the embodiments set forth herein, and it is intended that the present inventive concepts cover the various combinations, the modifications and variations of this invention without departing from the spirit and scope of the present inventive concepts. The appended claims should be construed to include other embodiments.

(76) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” “top,” “bottom,” and the like, may be used herein for ease of description to describe one element's or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(77) Ordinal numbers such as “first,” “second,” “third,” etc. may be used simply as labels of certain elements, steps, etc., to distinguish such elements, steps, etc. from one another. Terms that are not described using “first,” “second,” etc., in the specification, may still be referred to as “first” or “second” in a claim. In addition, a term that is referenced with a particular ordinal number (e.g., “first” in a particular claim) may be described elsewhere with a different ordinal number (e.g., “second” in the specification or another claim).

Claims

1. A semiconductor package, comprising: a substrate that includes an upper protection layer and a plurality of upper bonding pads; a semiconductor chip on the substrate; and a plurality of bonding wires connected to the semiconductor chip and the upper bonding pads, wherein each of the upper bonding pads includes: a first conductive pattern; a second conductive pattern that covers a top surface and a sidewall of the first conductive pattern and includes a metal element the same as a metal element of the first conductive pattern; and a bonding layer on the second conductive pattern, wherein a width of the top surface of the first conductive pattern is less than a width of a bottom surface of the first conductive pattern, and wherein the upper protection layer covers sidewalls of the second conductive pattern.
2. The semiconductor package of claim 1, wherein a width of a top surface of the second conductive pattern is about 90% to about 110% of the width of the bottom surface of the first conductive pattern.
3. The semiconductor package of claim 1, wherein each of the upper bonding pads further includes an intermediate layer between the second conductive pattern and the bonding layer, wherein the intermediate layer includes a metal element different from the metal element of the second conductive pattern and different from a metal element of the bonding layer.
4. The semiconductor package of claim 3, wherein the bonding layer covers a top surface and a sidewall of the intermediate layer.
5. The semiconductor package of claim 1, wherein a width of the bonding layer is the same as or greater than a width of a top surface of the second conductive pattern.
6. The semiconductor package of claim 5, wherein a pitch of the upper bonding pads is in a range of about 40 μm to about 60 μm , and the width of the bonding layer is in a range of about 20 μm to about 33 μm .
7. The semiconductor package of claim 1, wherein the bonding layer extends onto a top surface of the upper protection layer.
8. The semiconductor package of claim 1, wherein a grain size of the second conductive pattern is

different from a grain size of the first conductive pattern.

9. The semiconductor package of claim 1, wherein the bonding layer includes a metal different from a metal of the first conductive pattern and different from a metal of the second conductive pattern.

10. The semiconductor package of claim 1, wherein each of the upper bonding pads further includes an intermediate layer between the second conductive pattern and the bonding layer, the first conductive pattern and the second conductive pattern include copper, the intermediate layer includes nickel, the bonding layer includes gold, and the upper protection layer includes a solder resist material.

11. The semiconductor package of claim 1, wherein the first conductive pattern and second conductive pattern are formed in openings in the upper protection layer.

12. A semiconductor package, comprising a substrate that includes an upper protection layer and an upper bonding pad, wherein the upper bonding pad includes: a first conductive pattern; a second conductive pattern on the first conductive pattern; and a bonding layer on the second conductive pattern, wherein a width of a top surface of the first conductive pattern is less than a width of a bottom surface of the first conductive pattern, wherein the second conductive pattern covers a sidewall and the top surface of the first conductive pattern, wherein the upper bonding pad further includes an intermediate layer on the first conductive pattern, wherein the bonding layer covers a top surface and a sidewall of the intermediate layer, and wherein the upper protection layer covers sidewalls of the second conductive pattern.

13. The semiconductor package of claim 12, wherein a width of the bonding layer is about 100% to about 120% of the width of the top surface of the second conductive pattern.

14. The semiconductor package of claim 12, further comprising: a semiconductor chip on a top surface of the substrate, the semiconductor chip including a chip pad; and a bonding wire electrically connected to the chip pad and the upper bonding pad, wherein the bonding wire contacts the bonding layer.

15. The semiconductor package of claim 12, wherein a width of a top surface of the second conductive pattern is about 90% to about 110% of the width of the bottom surface of the first conductive pattern.

16. The semiconductor package of claim 12, wherein the second conductive pattern includes a metal element the same as a metal element of the first conductive pattern, and the intermediate layer includes a metal element different from the metal element of the second conductive pattern and different from a metal element included in the bonding layer.

17. A semiconductor package, comprising: a substrate, including: a dielectric structure that includes a plurality of stacked dielectric layers, a wiring line between the dielectric layers, a lower pad disposed on a bottom surface of the dielectric structure and electrically connected to the wiring line, an upper bonding pad disposed on a top surface of the dielectric structure and electrically connected to the wiring line, and an upper protection layer on the top surface of the dielectric structure; a solder ball on a bottom surface of the lower pad; a semiconductor chip on a top surface of the substrate; a bonding wire connected to the semiconductor chip and the upper bonding pad; and a molding layer on the substrate, the molding layer covering the semiconductor chip and the bonding wire, wherein the upper bonding pad includes: a first conductive pattern; a second conductive pattern on a top surface of the first conductive pattern, the second conductive pattern including a first metal element the same as a metal element of the first conductive pattern; an intermediate layer on a top surface of the second conductive pattern, the intermediate layer including a second metal element different from the first metal element; and a bonding layer on a top surface of the intermediate layer, the bonding layer including a third metal element different from the first metal element and different from the second metal element, wherein a width of the top surface of the first conductive pattern is less than a width of a bottom surface of the first conductive pattern, and wherein the upper protection layer covers a sidewall of the second

conductive pattern.

18. The semiconductor package of claim 17, wherein the second conductive pattern covers a sidewall of the first conductive pattern.

19. The semiconductor package of claim 17, wherein a width of the top surface of the second conductive pattern is about 90% to about 110% of the width of the bottom surface of the first conductive pattern, and a width of the bonding layer is the same as or greater than the width at the top surface of the second conductive pattern.

20. The semiconductor package of claim 17, wherein the upper bonding pad is one of a plurality of bonding pads that are laterally spaced apart from each other, an interval between the upper bonding pads is in a range of about 10 μm to about 20 μm , the width of the bottom surface of the first conductive pattern is in a range of about 17 μm to about 30 μm , and a width of the bonding layer is in a range of about 20 μm to about 33 μm .
